

Assignment on
Function Point Analysis of SPL-2 Project

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Chapter 1: Project Overview

1.1 Proposed System

Proposed System for PMIS

Account Creation: The admin will create departments and set a director for each department. The director of each department will add researchers. The admin and director will set an email address and a password for creating an account. The password can be changed anytime.

Login: Admin and researcher must go through an authorized login system to access his account. He/she has to enter his Gmail account and a password to log in. In case of password forgetting the members can set a new password with the help of code verification sent to their mail address.

Edit/Set Information: At the first login, the researcher will be asked to set up his/her profile. The researcher will be asked about a good amount of information in some categories. The main goal is not to bother the researcher time by time by asking for information. However, he/she can edit or update any provided information at any time. He/She has to enter the following information:

- ❖ Personal Information (list form)
- ❖ Contact Information (list form)
- ❖ Joining Information (list form)
- ❖ Spouse and Children Information
- ❖ Spouse Information (Tabular form)
- ❖ Children's Information (Tabular form)
- ❖ Education Information (Tabular form)
- ❖ Training Information (tabular form)
- ❖ Posting Information (tabular form)
- ❖ Promotion Information (tabular form)
- ❖ Achievement
- ❖ Conference Proceedings(tabular form)
- ❖ Developed Process(tabular form)

- ❖ Patent(tabular form)
- ❖ Books/Book Chapters(tabular form)
- ❖ Award/Grant(tabular form)
- ❖ Leave/Deputation Information(tabular form)
- ❖ Former/Other Service(tabular form)
- ❖ Other Activities/Assignment(tabular form)
- ❖ Disciplinary Information(tabular form)

General viewer: A general viewer is a person outside the organization. This viewer can only view the website, visit profiles or departments and search for any researcher. The organization members will also be treated as general viewers if he/she is not logged in.

Organized and user-friendly: All the information will be shown in an organized way with the best-experienced user interface.

Proposed System for R&D management

Research call: The Research Coordinator Head(Project Inspector) is the head of the research coordinator. He/she will declare a project call for a particular season with a deadline to submit a project proposal through his profile. The Research Coordinator Head can edit the deadline for submitting the project at any time.

Submit Proposal: The researcher can then fill out the proposal form through the website. He has to enter the following information

- ❖ Name of the unit
- ❖ Title of the proposed R & D project
- ❖ Name and designation of Project Leader & Researcher associates
- ❖ Type/ Nature of the proposed R & D project
- ❖ Introduction/ Background of the proposed R & D project
- ❖ Objective
- ❖ Socio-economic importance of the project
- ❖ Previous Professional Training/ Experience of the project leader relevant to the proposed R&D Project:
- ❖ Work Plan/ Work Packages of the proposed R&D Project:
- ❖ The expected outcome of the project
- ❖ Quarterly time-bound action plan for the proposed project
- ❖ Research facilities available in BCSIR
- ❖ Previous Professional Training/ Experience of the project leader relevant to the proposed R&D Project:
- ❖ A list of facilities (Equipment/Instrument) will be required in addition to the implementation of the R&D projects.

- ❖ What other R & D program in the related field has already been implemented in the BCSIR or elsewhere in the Country or Abroad?
- ❖ Implementation period
- ❖ Budget Information
- ❖ Name and Progress of another ongoing project of the project (As Project leader/ Associate):
- ❖ Any other Information Relevant to the R&D Project Proposal and its Execution
- ❖ Signature of the Project Leader
- ❖ Signature of the Head of the Unit

RD Head Approval: The submitted proposal will go next to the RD head for approval. The RD head can approve the project proposal by checking the research proposal. The RD Head can also deny proposals. In that case the proposal will be deleted.he/she has to approve or deny a proposal by confirming with his/her password

Director Approval: The proposal will go next to the director for approval. The director can approve the project proposal by checking the research proposal. The director can also deny proposals. In that case the proposal will be deleted.he/she has to approve or deny a proposal by confirming with his/her password

Proceed to the Committee: Those proposals which are approved by the director will go to the database for the committee's review. After the deadline, the committee will generate a sheet from all the proposals. It will help the committee to make decisions efficiently.

Monitoring: The research coordinator committee will monitor the ongoing project. There is a maintenance page on the website and the committee comments on this page and updates project work. All projects are tracked on websites.

1.2 QFD

Normal Requirement:

- ❖ Web-based application.
- ❖ Anyone can view researchers' public data.
- ❖ Admin can add Research Coordinator Head.
- ❖ Call for a research proposal by the Research Coordinator Head.
- ❖ The researcher will fill up the proposal form after a proposal call.
- ❖ The RD Head can approve the submitted proposal for further process or deny the proposal.

- ❖ Director can approve the proposal for further process or deny the proposal forwarded from the RD head.
- ❖ The Research Coordinator Head can finalize the proposal or reject the proposal forwarded from the Director.
- ❖ Search people easily and add a filter in the search.
- ❖ View notices provided by the organization.
- ❖ Edit their personal information anytime.
- ❖ Admin, Director can add a researcher.
- ❖ Add their personal information at the time of first login.

Expected Requirements:

- ❖ Organized profile.
- ❖ Admin, Director can remove a researcher.
- ❖ Responsiveness application
- ❖ The proposal form will contain personal information automatically at the time of filling up the proposal form.
- ❖ Members are informed or notified by Email or dashboard or both.
- ❖ The Research Coordinator Head can monitor ongoing projects.

Exciting requirements:

- ❖ Connection graph of each researcher.
- ❖ Advanced search functionality to easily find specific researchers.

Chapter 2: Work Description

2.1 Function Point Analysis (FPA)

To get FPA we need to calculate UFP (Unadjusted Function Points) and TCF (Technical Complexity Factor). Because,

$$FPA = UFP * TCF$$

2.1.1 Calculating UFP

To get UFP, we need to consider a table of FPA Matrices with weights.

Item	Simple	Average	Complex
External Inputs (EI)	3	4	6
External Outputs (EO)	4	5	7
External Inquiries (EQ)	3	4	6
External Interface Files (EIF)	7	10	15
Internal Logical Files (ILF)	5	7	10

In the Table:

i. Identify External Inputs (EI):

External Inputs are user interactions that provide data to the system. Identify and count all the different inputs the system receives from external sources. Examples include user inputs, sensor data, and data from other systems. Examples: User inputs (like filling forms), sensor data, or data from other software.

ii. Identify External Outputs (EO):

External Outputs are user interactions where the system sends out data. Identify and count all the different outputs the system generates for external entities. Examples include reports, messages, and data sent to other systems. Examples: Reports, messages, or data sent to other software.

iii. Identify External Inquiries (EQ):

External Inquiries are interactions where the user gives input, and the system provides immediate output. Count the number of different inquiries the system processes. Examples include online data inquiries and queries. Examples: Online data inquiries or queries where users ask for specific information.

iv. Identify Internal Logical Files (ILF):

Internal Logical Files are logical groups of data used and maintained by the application. Identify and count all unique data groups within the system that are used internally. Examples include files, databases, and data structures. Examples: Files, databases, or data structures used within a software.

v. Identify External Interface Files (EIF):

External Interface Files are data groups that are used and maintained by external applications but are referenced by the current application. Identify and count all unique data groups that are used by your application but are maintained by external applications. Examples: Data groups maintained by other software that your system references.

Each of these items has its own weight based on the complexity: Simple, Average, Complex.

$$UFP = (EI \times \text{Weight}_{EI}) + (EO \times \text{Weight}_{EO}) + (EQ \times \text{Weight}_{EQ}) + (ILF \times \text{Weight}_{ILF}) + (EIF \times \text{Weight}_{EIF})$$

2.1.2 Calculating TCF

Technical Complexity Factors: Each component or subfactor is rated from 0 to 5, where 0 means the subfactor is irrelevant, 3 means it is average, and 5 means it is essential to the system being built.

Ratings	
0	Not present / No influence
1	Insignificant
2	Moderate
3	Average
4	Significant
5	Essential

Factors:

- I. Data Communication
- II. Distributed Functions
- III. Performance
- IV. Heavily Utilized Hardware
- V. High Transaction Rates
- VI. Online Data Entry
- VII. Online Updating
- VIII. End-user Efficiency
- IX. Complex Computations
- X. Reusability

- XI. Ease of Installation
- XII. Ease of Operation
- XIII. Portability
- XIV. Maintainability/Facility change

2.1.3 Calculating FP

$$FP = UFP \times TCF$$

Identifying Language Factor:

Details list of LOC against FP for each language is provided here:

<https://www.qsm.com/resources/function-point-languages-table>

2.2 Using Cocomo Model to Estimate Cost

Tasks in this Step:

- ❖ Using COCOMO II online Calculator. It can be found here:
<http://softwarecost.org/tools/COCOMO/>
- ❖ Modification of Software Scale Drivers, Software Cost Drivers factors according to project
- ❖ Result Analysis

Chapter 3: Function Point Calculation

3.1 Identifying FPA Matrix

For our described project the FPA metrics are:

For PMIS:

i. External Inputs (EI):

- ❖ Admin creating departments.
- ❖ Director adding researchers.
- ❖ Admin and director setting up email addresses and passwords.
- ❖ Researchers updating personal, contact, joining, spouse and children, education, training, posting, promotion, achievement, conference proceedings, developed processes, patents, books/chapters, awards/grants, leave/deputation, former/other service, disciplinary information, and other activities/assignments information.
- ❖ Admin Setting department head
- ❖ User can change their password
- ❖ User can edit their profile
- ❖ Viewer search for an information

ii. External Outputs (EO):

- ❖ Notifications sent to researchers for password reset.
- ❖ Verification code sending
- ❖ Researchers view their own profile
- ❖ Profiles of researchers and departments displayed to general viewers.
- ❖ Website displaying organized information about researchers and departments.

iii. External Inquiries (EQ):

- ❖ General viewers searching for researchers
- ❖ Searching for departments.
- ❖ Searching for RD head

iv. Internal Logical Files (ILF):

- ❖ Information about departments.

- ❖ Information about researchers
- ❖ Information about committee
- ❖ Information about projects
- ❖ Research proposals and associated information.
- ❖ Education information
- ❖ Achievement information

v. External Interface Files (EIF):

- ❖ Predefined or present organization status
- ❖ Email System

For R&D Management System:

i. External Inputs (EI):

- ❖ Research Coordinator Head declaring project calls.
- ❖ Researchers filling out project proposal forms, including project details, work plan, budget, etc.
- ❖ RD Head and Director approving or denying project proposals.
- ❖ Committee generating review sheets from project proposals.

ii. External Outputs (EO):

- ❖ Notifications to researchers about proposal approvals or denials.
- ❖ Committee comments and project updates displayed on the maintenance page.
- ❖ Review sheets generated for committee decision-making.

iii. External Inquiries (EQ):

- ❖ Researchers inquired about the status of their project proposals.
- ❖ Researchers inquiries while selecting teammates.

iv. Internal Logical Files (ILF):

- ❖ Project proposal information (name, title, objectives, budget, etc.).
- ❖ Approval status of project proposals.
- ❖ Committee comments and project progress updates.

v. External Interface Files (EIF):

- ❖ Personal info setting while submitting project proposal

Item	Count	Weight	Count*Weight
External Inputs (EI)	12	4	48
External Outputs (EO)	8	5	40
External Inquiries (EQ)	5	4	20
External Interface Files (EIF)	3	10	30
Internal Logical Files (ILF)	10	7	70
Total:			208

So, Our UFP = 208

3.2 Identifying TCF

Item	Ratings
Data Communication	4
Distributed Functions	0
Performance	3
Heavily Utilized Hardware	1
High Transaction Rates	1
Online Data Entry	5
Online Updating	5
End-user Efficiency	5
Complex Computations	3
Reusability	5
Ease of Installation	5
Ease of Operation	4
Portability	2
Maintainability/Facility change	3
Total:	47

$$\text{So, TCF} = 0.65 + (0.01 \times 47) \\ = 1.12$$

3.3 Calculating Function Point (FP) & LOC

As we know, $\text{FP} = \text{UFP} \times \text{TCF}$

Here, in our project UFP value is: 208
TCF Value is: 1.12

$$\text{So, FP} = 208 \times 1.12 \\ = 232.96$$

Estimation Language of our project is:

Javascript	95%
HTML	5%

LOC Calculation:

Language	Use in Project	Associated FP	Language Factor	LOC
javascript	95%	221.312	47	10401.664
HTML	5%	11.648	34	396.032
Total:				10797.696 ≈ 10798

Chapter 4: Cost Estimation

4.1 Defined/Modified Parameters

- ❖ Sizing Method : LOC
- ❖ New Lines : 10798
- ❖ Development Flexibility : Low
- ❖ Architecture : High
- ❖ Data Base Size : High
- ❖ Platform Experience : Low
- ❖ Time Constraint : Very High

4.2 Cost Calculation using COCOMO II Calculator

COCOMO II - Constructive Cost Model

Monte Carlo Risk
Auto Calculate

Software Size

Sizing Method

SLOC

% Design Modified

% Code Modified

% Integration Required

Assessment and Assimilation (0% - 8%)

Software Understanding (0% - 50%)

Unfamiliarity (0-1)

New

Reused

Modified

Software Scale Drivers

Precedentedness

Architecture / Risk Resolution

Process Maturity

Development Flexibility

Team Cohesion

Software Cost Drivers

Product

Required Software Reliability

Data Base Size

Product Complexity

Personnel

Analyst Capability

Programmer Capability

Platform

Time Constraint

Storage Constraint

Platform Volatility

Software Development (Elaboration and Construction)

Effort = 71.7 Person-months

Schedule = 14.2 Months

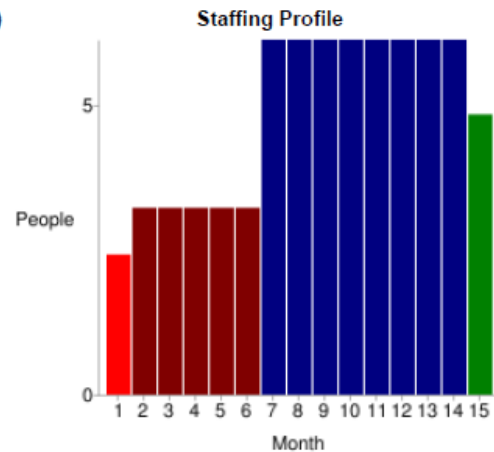
Cost = \$1434

Total Equivalent Size = 10798 SLOC

Effort Adjustment Factor (EAF) = 1.80

Acquisition Phase Distribution

Phase	Effort (Person- months)	Schedule (Months)	Average Staff	Cost (Dollars)
Inception	4.3	1.8	2.4	\$86
Elaboration	17.2	5.3	3.2	\$344
Construction	54.5	8.9	6.1	\$1091
Transition	8.6	1.8	4.8	\$172



Software Effort Distribution for RUP/MBASE (Person-Months)

Phase/Activity	Inception	Elaboration	Construction	Transition
Management	0.6	2.1	5.5	1.2
Environment/CM	0.4	1.4	2.7	0.4
Requirements	1.6	3.1	4.4	0.3
Design	0.8	6.2	8.7	0.3
Implementation	0.3	2.2	18.5	1.6
Assessment	0.3	1.7	13.1	2.1
Deployment	0.1	0.5	1.6	2.6

Developed for
Reusability

Nominal ▼

Documentation Match to
Lifecycle Needs

Nominal ▼

Personnel
Continuity

Nominal ▼

Application
Experience

Nominal ▼

Platform
Experience

Low ▼

Language and
Toolset Experience

Nominal ▼

Project

Use of Software
Tools

Nominal ▼

Multisite
Development

Nominal ▼

Required
Development
Schedule

Nominal ▼

Maintenance ▼

Software Labor Rates

Cost per Person-Month (Dollars)

Results

4.3 Results

Software Development (Elaboration and Construction)

Effort = 71.7 Person-months

Schedule = 14.2 Months

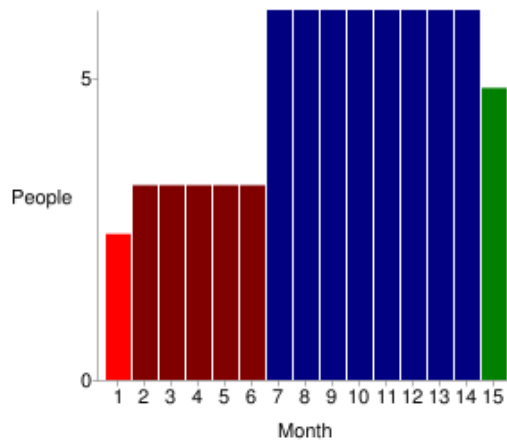
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Staffing Profile



Software Effort Distribution for RUP/MBASE (Person-Months)

Phase/Activity	Inception	Elaboration	Construction	Transition
Management	0.6	2.1	5.5	1.2

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Requirements	1.6	3.1	4.4	0.3
Design	0.8	6.2	8.7	0.3
Implementation	0.3	2.2	18.5	1.6
Assessment	0.3	1.7	13.1	2.1
Deployment	0.1	0.5	1.6	2.6

Chapter 5: Conclusion

In conclusion, this document outlines the proposed Project Management Information System (PMIS) and Research & Development Management System, providing a streamlined approach for personnel and project management. The systems incorporate user-friendly interfaces, systematic workflows, and real-time notifications, enhancing efficiency and accountability.

Through Function Point Analysis and the COCOMO II model, the document offers valuable insights into project complexity and resource requirements. The effort distribution across different phases and activities provides a clear roadmap for implementation.

These systems represent a significant leap toward digital transformation, fostering collaboration, transparency, and productivity within the organization. As technology evolves, these systems provide a robust foundation for future enhancements, ensuring the organization's agility and responsiveness to changing demands.

By embracing these innovative solutions, the organization can revolutionize its project and personnel management practices, driving success and efficiency. Continuous engagement and strategic implementation will be key to realizing the full potential of these systems, propelling the organization to new heights of excellence.